BETTER BASELINE / FRONTEND DESIGN SPECIFICATION

Dark instrument.
One workspace, sixteen states.

Approved visual direction / Slate work surface, mint accents, square controls, Inter labels and monospaced quantities. Every desktop and 375px mobile frame represents route / or an expanded section within it. No new pages, mode toggle, patient uploads, accounts or saved-project workflow.

Canvas · Working surface · Enrollment lock · Action / selection · Stale / warning · Invalid / error

Planning & recovery

01 Initial LIVE / no current design; Baseline collapse selected

02 Manual edit / alternate ARR input

03 Invalid ARR ≥ baseline / specific field feedback

04 Calculate & Lock in progress / preserved inputs

05 Current design / separate arm summary and immutable lock

13 Backend unavailable / retry with preserved inputs

14 DEMO / explicit illustrative results and blocked live actions

15 Reopened JSON / detached valid and invalid feedback

Stress, simulation & audit

06 Stress comparison / analytic curve and sensitivity grid

07 Stress changed / stale → loading, enrolled counts unchanged

08 Valid selected cell / selected-scenario simulation ready

09 Invalid null / selection and simulation blocked

10 Simulation running / alternate failed retry state

11 Current complete output schema / export eligible state

12 Stale simulation / setting mismatch and invalid settings

16 Mixed synthetic/manual provenance / methods & limitations

HOW TO READ THE MOCKUPS

‹Angle-bracket values› are service-return bindings, not fabricated results. Supplied fixture numbers (903 enrollment, ≈47% stress power) and drawn chart traces are labelled illustrative/ schematic. LIVE execution status does not turn a fixture into a current live result. No Monte Carlo outcome is invented.

Planning fields use percentages / percentage points; RR is a decimal ratio. Stress numeric fields use fractions with explicit conversions. Stress inherits the planned effect. Lock changes only on successful design calculation. Analytics are current only for matching inputs; a planned point at selected dropout is not necessarily the original plan.

Sliders: keyboard arrows, 1-point steps, labelled numeric alternatives. Valid chart/table cells: keyboard selection, visible focus, 44px hit targets, risk/loss/power announcements. Invalid cells: null × with reason, never 0%. Tables: captions and row/column headers. Status: visible words plus polite announcements. Invalid submission: first invalid field receives focus.

Complete JSON requires all displayed inputs and current design, curve/grid, selected simulation, seed and replicates to match. Grid CSV is independent. JSON validation checks a detached artifact; never restores assumptions or reruns calculations. DEMO cannot simulate or complete-export.

REVIEW CHECKLIST · No collapsed layouts · No clipped visible content · Readable contrast · Consistent alignment / spacing · Units and states explicit · Numerical chart alternatives available · Current/stale identity preserved

---

VALIDATION / NUMERICAL UNITS

Planning: risk, alpha and target strictly between 0% and 100%; target must exceed alpha. RR strictly between 0 and 1. ARR must be positive percentage points and less than baseline risk. Shared dropout is ≥0% and <100%.

Stress sliders: risk 1–99%, dropout 0–50%, steps 1 percentage point; numeric alternatives are fractions. Seed: integer 0–4,294,967,295. Replicates: integer 1–100,000. Defaults: 42 and 10,000.

IMPLEMENTATION / 375PX

The same workflow.
A narrower instrument.

All sixteen states have a narrow-screen counterpart. No mobile-only routes, model controls or enrollment overrides.

01–05 / Plan

Stack labeled percentage fields. Effect selection swaps RR (ratio) with ARR (percentage points). Keep validation at the field; submission focuses the first invalid input. A successful calculation alone replaces the lock.

06–09 / Stress

Keep frozen counts visible above risk/dropout controls. Sliders have fraction-entry alternatives (0.10 = 10%). Compare scenarios with explicit labels. Curve and heatmap have numerical selection tables; invalid/null cells stay unselectable.

10–12 / Simulate

Show one selected scenario, seed and replicates. Separate two-sided and favorable-direction results and each MCSE in pp. Expected retained counts are not sampled actual retained counts. Stale results never enable export.

13–16 / Recover & audit

Preserve inputs on service failure. DEMO cannot simulate or complete-export. JSON validation checks a detached artifact, never restores the workspace. Stack field provenance and scientific disclosures.

44px interaction targets · visible focus · text labels for current, loading, stale, invalid and unavailable · scoped table headers · status announcements.

Power is conditional on assumptions and methods. It is not a drug-efficacy or trial-success prediction.

VALIDATION / NUMERICAL UNITS

Planning: risk, alpha and target strictly between 0% and 100%; target must exceed alpha. RR strictly between 0 and 1. ARR must be positive percentage points and less than baseline risk. Shared dropout is ≥0% and <100%.

Stress sliders: risk 1–99%, dropout 0–50%, steps 1 percentage point; numeric alternatives are fractions. Seed: integer 0–4,294,967,295. Replicates: integer 1–100,000. Defaults: 42 and 10,000.